

EXAMPLE HARDWARE CONFIGURATIONS AND MODULAR SYSTEM VARIANTS

Purpose of Hardware Variant Disclosure

This section provides illustrative examples of hardware configurations and modular variants for the longitudinal penile physiology monitoring device. The examples are non-limiting and are intended to demonstrate that the inventive concepts may be implemented across multiple cost tiers, sensing capabilities, and structural form factors.

Variant A — Cost-Optimized Monitoring Device

A cost-optimized embodiment may include a minimal sensing set sufficient to infer physiologic activation using one or more deformation measurement modalities (e.g., strain, pressure, or magnetic displacement) combined with basic motion context. Wireless communication may operate in batch or periodic sync mode to conserve energy. This embodiment may prioritize simplicity, small profile, and extended battery life.

Variant B — Balanced Consumer Device

A balanced consumer embodiment may include multiple deformation sensing modalities with optional temperature sensing and advanced inertial context. The device may support multi-mode sampling including performance mode and longevity mode. Derived indices and confidence scoring may be generated locally or via companion computational systems.

Variant C — High-Fidelity Performance Device

A high-fidelity embodiment may include multi-modal sensing including deformation sensing, temperature arrays, perfusion sensing, advanced inertial measurement, and increased processing capability. This embodiment may support higher sampling rates, enhanced signal processing, and richer physiologic analytics including advanced rigidity modeling and detailed episode segmentation.

Variant D — Clinical or Research-Oriented Device

A clinical or research embodiment may incorporate expanded sensing modalities, calibrated measurement subsystems, enhanced data logging, and secure export features for research or clinical analysis. Such embodiments may support multi-user studies, longitudinal cohorts, or advanced physiologic modeling without limiting the invention to any regulatory classification.

Modular and Upgradeable Architectures

Embodiments may include modular components allowing upgrade of sensor modules, communication modules, power systems, or firmware. Modular architectures may enable the same base device body to support multiple tiers through replaceable sensing cartridges, optional add-on modules, or future component substitutions.

System Variant Combinations

The above variants may be deployed as standalone products or within multi-node systems. Different tiers may be mixed across nodes, allowing a cost-optimized Node 1 to operate with high-fidelity companion nodes or vice versa. The invention encompasses any such combination and subcombination of device tiers and modular configurations.

The hardware configuration embodiments described herein are illustrative and non-limiting and encompass present and future modular system variants capable of implementing longitudinal physiologic monitoring through wearable devices.